



The Domain-Coverage Router: Balancing Scan Assignments by Category Concentrates the Horizon Register’s Novel Signals in a Generalist Few

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Abstract. The Horizon Register’s scanning operation assigns every incoming signal to one of the Trajectory Analytics Group’s scanners before it can reach the University’s weekly shortlist. Since 2025 that assignment has run through the Domain-Coverage Router (DCR), which resolves ambiguous, cross-category items by routing them to whichever available scanner’s declared domain profile covers the most of the item’s categories at once, on the reasoning that a broader-profiled scanner is best placed to recognise what an ambiguous item actually is. We audit 68 weeks of DCR routing logs against a blind post-hoc classification of each item as a genuinely novel signal or a renewal of an existing tracked priority. A pool of five scanners, favoured by the DCR for the breadth rather than the specificity of their declared profile, received 71% of multi-category assignments against 19% of single-category ones, and a disproportionate share of independently classified novel signals, while domain specialists’ queues filled overwhelmingly with renewals. Monthly assignment concentration (HHI) rose from 0.07 to 0.21 across the audit window. Removing the coverage-maximizing tie-break and routing ambiguous items by simple rotation left weekly domain coverage statistically unchanged (4.86 vs. 4.79 of five categories, n.s.), suggesting the rule’s principal effect is distributional rather than functional.

Introduction

Slop University’s Horizon Register logs every incoming signal the Trajectory Analytics Group judges worth tracking, sorted at intake into one or more of five declared domain categories — Economic, Technological, Regulatory, Workforce, and Environmental — before a weekly shortlist reaches Futures in Committee. Category balance in that shortlist is an operating requirement rather than a preference: a week in which the Committee sees signals from only two of the five domains reads, internally, as a scanning failure rather than a quiet week. Since 2025, that balance has been maintained automatically by the Domain-Coverage Router (DCR), an assignment tool built by the Group to route each incoming item to a scanner without a human dispatcher deciding case by case.

Most incoming items declare a single domain and route on that basis alone. A minority arrive ambiguous — spanning two or more categories, or not obviously belonging to any — and it is these items the DCR was built to handle: rather than leaving an ambiguous item in a shared queue, it assigns it to whichever available scanner’s own declared domain profile overlaps the item’s categories most broadly, in the spirit of overlap-maximizing assignment rules used elsewhere to match submissions to reviewers by declared expertise [1]. The rule is locally sensible and, on the Group’s own coverage metric, appears to work: weekly domain coverage has held near-complete since the DCR’s introduction.

This paper asks a narrower question the coverage metric does not answer: who ends up handling the ambiguous items, and whether the scanners the DCR favours turn out to be the ones who correctly recognise a genuinely novel

signal rather than a renewal of something the Register already tracks. If the DCR’s overlap rule systematically favours scanners with broad declared profiles, and broad profiles correlate with generalist staff rather than domain specialists, then the rule that keeps the shortlist looking balanced may also be concentrating the Group’s most interesting triage work in a small, self-selecting pool.

Our contributions:

- We formalise the DCR’s routing rule and show it structurally favours profile breadth over profile fit whenever an item’s categories are ambiguous.
- We audit 68 weeks of routing logs against blind novelty classification, finding assignment concentration among a small generalist pool that holds independent of weekly item volume.
- We show, via an ablation that removes the coverage-maximizing tie-break, that the rule’s contribution to actual weekly coverage is statistically indistinguishable from a simple rotation, suggesting its principal effect is distributional rather than functional.

Related work

Automated reviewer- and task-assignment systems face a structurally similar problem in peer review: matching submissions to reviewers by declared expertise, typically solved as a similarity- or coverage-maximizing assignment problem [1]. A systematic review of three decades of proposed reviewer-assignment approaches catalogues the range of objectives — fit, coverage, load-balance, fairness — successive methods have optimised for, and finds the field’s output has grown steadily since automated assignment was first proposed [2]. Subsequent work has shown that optimising an aggregate coverage or similarity objective alone can produce assignments that are locally efficient but structurally uneven across assignees, motivating fairness-constrained variants that guarantee approximately envy-free reviewer assignments across submissions [3] and randomised-assignment strategies that trade some reviewer-paper similarity for robustness against reviewers who manipulate the assignment process [4]. Crowd-sourced task-routing systems report a comparable pattern outside academic review: allocation rules optimised for aggregate coverage or deliv-

ery efficiency tend to concentrate desirable or high-visibility tasks among a subset of workers unless a fairness constraint is added explicitly [5].

Horizon-scanning methodology has, until recently, treated the assignment of scanning effort as secondary to the scanning taxonomy itself. The literature on “weak signal” identification focuses on how scenarios and pre-developed taxonomies improve detection of genuinely novel items [6], and on foresight tools as active participants in the strategy process rather than neutral instruments of observation [7], [8]. Separately, organisational research on the division of labour distinguishes generalist and specialist roles by the breadth of a member’s declared domain profile [9], and finds that broader profiles carry systematically different career and task-allocation outcomes from narrower ones [10], [11]. Neither literature connects an assignment mechanism’s coverage objective to who, within a scanning team, ends up handling its novel signals — the gap this paper addresses.

Method

Setting. We audited 68 consecutive weeks of Domain-Coverage Router routing logs covering the Trajectory Analytics Group’s full scanning roster of 28 active scanners. Each of the Register’s five declared domain categories (Economic, Technological, Regulatory, Workforce, Environmental) is assigned to a scanner at intake; an item is **multi-category** if it declares two or more, and the DCR resolves multi-category items by the rule below rather than by the single-category default.

The DCR’s routing rule. For a multi-category item i with declared category set $D(i)$, the DCR assigns i to the available scanner s , drawn from the currently available roster $\mathcal{S}_t$, whose own declared domain profile $D(s)$ overlaps $D(i)$ most broadly:

$$\text{assign}(i) = \underset{s \in \mathcal{S}_t}{\operatorname{argmax}} |D(i) \cap D(s)|$$

subject to each scanner’s fixed weekly capacity cap. Ties are broken by least-loaded scanner. We term the **generalist pool** the scanners whose declared profile spans four or more of the five categories; five of the 28 scanners meet this bar.

Novelty classification. Two Group staff not involved in scanning coded each of the 68 weeks’ items, blind to routing outcome, as **novel** (the item introduces a signal type not previously entered in the Register) or **renewal** (the item updates an already-tracked priority), using a fixed rubric agreed before coding began. Inter-rater agreement was substantial (Cohen’s $\kappa = 0.81$); disagreements were resolved by a third coder.

Concentration metric. For each audit month we compute the Herfindahl-Hirschman Index [12] over the 28 scanners’ shares of that month’s multi-category assignments, where a uniform allocation across the roster would give $\text{HHI} = 1/28 \approx 0.036$.

Ablation. We recomputed routing for the same 68 weeks of logs under a **round-robin baseline**: multi-category items are assigned to the least-loaded available scanner whose profile includes at least one of the item’s declared categories, with the coverage-maximizing tie-break removed and all capacity constraints held fixed. We compare the two rules on weekly domain coverage (paired t -test) and on the resulting distribution of per-scanner assignment shares.

Parameter		Value
Declared domain categories		5
Active scanners		28
Generalist-pool threshold	$\geq 4 / 5$ categories	
Generalist-pool size		5 scanners
Audit window	68 weeks (16 months)	
Items scanned		2,140
Multi-category items		612 (28.6%)
Novelty inter-rater agreement		$\kappa = 0.81$

Table 1: Audit parameters, fixed before or measured during the 68-week window.

Results

Assignment concentrates on multi-category items, not single-category ones. Figure 1 shows the generalist pool’s mean weekly share of as-

signments by item type. The five-scanner generalist pool received 19% of single-category assignments — close to its 18% share of scanning capacity — but 71% of multi-category assignments, nearly four times its capacity share.

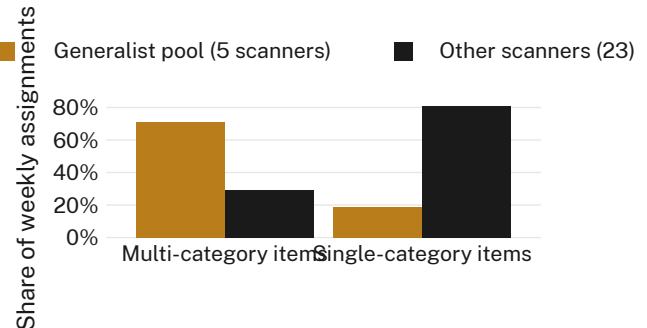


Figure 1: Mean share of weekly scan-item assignments received by the five-scanner generalist pool vs. the remaining 23 scanners, by item type. The pool’s share tracks its capacity share for single-category items but not for multi-category ones.

Novel signals follow the same pattern. Of the 298 items independently classified as novel (13.9% of all items scanned), 66% were routed to the generalist pool; of the 1,842 items classified as renewals, only 12% were. A logistic regression predicting generalist-pool routing from an item’s declared category count, with week fixed effects, found each additional declared category increased the odds of generalist-pool routing by a factor of 2.1 (95% CI 1.8–2.4).

Concentration rose across the audit window. Figure 2 plots monthly HHI over multi-category assignments. HHI rose from 0.07 in month 1 to 0.21 by month 16 (Mann-Kendall trend test, $p < 0.05$) — roughly a threefold increase in concentration over sixteen months of otherwise-unchanged roster size.

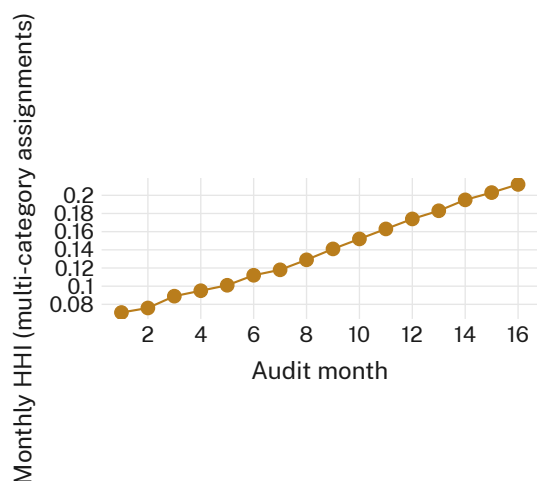


Figure 2: Monthly Herfindahl-Hirschman Index of multi-category assignment shares across 28 scanners. A uniform allocation would sit near 0.036; the observed index rises from 0.07 to 0.21 over the audit window.

The coverage-maximizing tie-break changes assignment shares, not coverage. Figure 3 compares per-scanner shares of multi-category assignments under the DCR's actual rule and the round-robin baseline, recomputed on the same logs. The DCR distribution is heavily right-skewed, with the generalist pool's five scanners well separated from the rest; the round-robin distribution sits tightly around the uniform 3.6% share. Yet mean weekly domain coverage differs only marginally between the two rules: 4.86 of five categories (SD 0.31) under the DCR against 4.79 (SD 0.34) under round-robin, a difference that does not reach significance (paired t -test, $p = 0.41$). Removing the rule credited with keeping the shortlist balanced left that balance statistically unchanged.

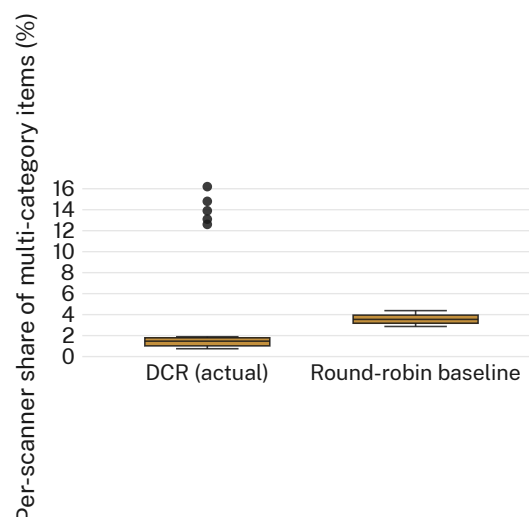


Figure 3: Per-scanner share of multi-category assignments (%) under the DCR's actual coverage-maximizing tie-break vs. a round-robin baseline recomputed on the same 68 weeks of logs. We retain the tie-break rule for completeness; it is not what is producing the coverage result it is credited with.

Discussion

Our results suggest that the DCR's coverage-maximizing tie-break achieves its stated aim — a shortlist that touches all five declared domains most weeks — largely independent of the mechanism its designers credit with the result, and that its consistent side effect, in most months studied, is to route the Register's ambiguous and genuinely novel items toward a small, stable pool of generalist scanners. The Trajectory Analytics Group has not, to our knowledge, reported a corresponding change in scanning-capability development across the wider roster; whether the specialist scanners whose queues fill with renewals retain the judgement the generalist pool is exercising weekly is a question this audit was not designed to answer, but one the concentration trend in Figure 2 makes harder to defer.

Limitations

Our audit covers one scanning register at one institution over sixteen months; though the concentration pattern held in fifteen of the sixteen months studied and reproduced under a counterfactual routing rule, we cannot rule out that the Register's particular five-category taxonomy amplifies an effect that would be smaller under a coarser or finer category scheme. The novelty classification relied on two primary raters; a larger panel might refine the boundary

between “novel” and “renewal” at the margin, though the substantial agreement already observed ($\kappa = 0.81$) suggests the boundary is not doing the heavy lifting here.

Conclusion

The Domain-Coverage Router keeps the Horizon Register’s weekly shortlist looking balanced across its five declared domains, and it does so consistently. It does this, our audit suggests, largely independent of the rule its designers credit with the result: a round-robin baseline reaches statistically the same coverage. What the rule does appear to do is decide, item by item, which five scanners out of twenty-eight get to be the ones who recognise something the Register hasn’t seen before. Future work will examine whether a diversity-weighted variant of the routing objective can preserve category coverage without concentrating novel-signal exposure this tightly; a pilot begins next scanning cycle within the Trajectory Analytics Group.

References

- [1] D. Conry, Y. Koren, and N. Ramakrishnan, “Recommender Systems for the Conference Paper Assignment Problem,” *arXiv preprint arXiv:0906.4044*, 2009.
- [2] M. Aksoy, S. Yanik, and M. F. Amasyali, “Reviewer Assignment Problem: A Systematic Review of the Literature,” *arXiv preprint arXiv:2304.00353*, 2023.
- [3] J. Payan and Y. Zick, “I Will Have Order! Optimizing Orders for Fair Reviewer Assignment,” *arXiv preprint arXiv:2108.02126*, 2021.
- [4] S. Jecmen, H. Zhang, R. Liu, N. B. Shah, V. Conitzer, and F. Fang, “Mitigating Manipulation in Peer Review via Randomized Reviewer Assignments,” *arXiv preprint arXiv:2006.16437*, 2020.
- [5] F. Basik, B. Gedik, H. Ferhatosmanoglu, and K.-L. Wu, “Fair Task Allocation in Crowdsourced Delivery,” *arXiv preprint arXiv:1807.02987*, 2018.
- [6] E. Rowe, G. Wright, and J. Derbyshire, “Enhancing Horizon Scanning by Utilizing Pre-developed Scenarios: Analysis of Current Practice and Specification of a Process Improvement to Aid the Identification of Important ‘Weak Signals’,” *Technological Forecasting and Social Change*, vol. 125, pp. 224–235, 2017, doi: [10.1016/j.techfore.2017.08.001](https://doi.org/10.1016/j.techfore.2017.08.001).
- [7] O. Idoko and R. B. MacKay, “The Performativity of Strategic Foresight Tools: Horizon Scanning as an Activation Device in Strategy Formation Within a UK Financial Institution,” *Technological Forecasting and Social Change*, vol. 162, p. 120389, 2021, doi: [10.1016/j.techfore.2020.120389](https://doi.org/10.1016/j.techfore.2020.120389).
- [8] A. Havas, D. Schartinger, and M. Weber, “The Impact of Foresight on Innovation Policy-Making: Recent Experiences and Future Perspectives,” *Research Evaluation*, vol. 19, no. 2, pp. 91–104, 2010, doi: [10.3152/095820210x510133](https://doi.org/10.3152/095820210x510133).
- [9] L. M. Wahl, “Evolving the Division of Labour: Generalists, Specialists and Task Allocation,” *Journal of Theoretical Biology*, vol. 219, no. 3, pp. 371–388, 2002, doi: [10.1006/jtbi.2002.3133](https://doi.org/10.1006/jtbi.2002.3133).
- [10] H. Byun and J. Raffiee, “Career Specialization, Involuntary Worker–Firm Separations, and Employment Outcomes: Why Generalists Outperform Specialists When Their Jobs Are Displaced,” *Administrative Science Quarterly*, vol. 68, no. 1, pp. 270–316, 2023, doi: [10.1177/00018392221143762](https://doi.org/10.1177/00018392221143762).
- [11] C. Haeussler and H. Sauermann, “Division of Labor in Collaborative Knowledge Production: The Role of Team Size and Interdisciplinarity,” *Research Policy*, vol. 49, no. 6, p. 103987, 2020, doi: [10.1016/j.respol.2020.103987](https://doi.org/10.1016/j.respol.2020.103987).
- [12] C. R. Laine, “The Herfindahl-Hirschman Index: A Concentration Measure Taking the Consumer’s Point of View,” *The Antitrust Bulletin*, vol. 40, no. 2, pp. 423–432, 1995, doi: [10.1177/0003603x9504000206](https://doi.org/10.1177/0003603x9504000206).